

Backward Chaining

Ex No: 7

Date: 9/9/2026

AIM

To study and implement unification and resolution exercises using Prolog facts, compound terms, queries, and rules.

Exercise 1: Medical Diagnosis

% Facts

has_fever(john).

has_cough(john).

% Rule

has_flu(X) :- has_fever(X), has_cough(X).

% Query

% ?- has_flu(john).

Output:

true.

Exercise 2: Student Eligibility Checker

% Facts

passed(alice, maths).

passed(alice, science).

% Rule

eligible(X) :- passed(X, maths), passed(X, science).

% Query

% ?- eligible(alice).

Output:

true.

Exercise 3: Travel Planner

% Facts

has_ticket(james).

has_visa(james).

% Rule

can_travel(X) :- has_ticket(X), has_visa(X).

% Query

% ?- can_travel(james).

Output:

True.

RESULT

Thus, the unification and resolution exercises were implemented successfully and the expected Prolog outputs were obtained.